Topic 6: Transport Layer

TCP, UDP, Flow Control, and Error Control

151. Which of the following protocols are examples of TCP/IP transport layer protocols? (Choose two)

A) HTTP and FTP   B) TCP and UDP   C) IP and DNS   D) SMTP and POP3

152. The Transport Layer is responsible for:

A) Logical addressing only   B) Session establishment, segmentation, sequencing, error detection, and flow control   C) Physical addressing only   D) Data representation

153. Segmentation is the process of:

A) Combining small parts into large data   B) Dividing data into small parts called segments   C) Encrypting data   D) Compressing data

154. Sequencing in the transport layer means:

A) Encrypting segments   B) Giving a serial number to each segment   C) Combining segments   D) Deleting duplicate segments

155. Error detection in the transport layer uses:

A) Parity check   B) CRC (Cyclic Redundancy Check)   C) Checksum only   D) Hash only

156. CRC stands for:

A) Code Read Check   B) Cyclic Read Check   C) Cyclic Redundancy Check   D) Code Redundancy Check

157. When a segment arrives at the receiver with an error, the receiver:

A) Ignores it   B) Drops it and orders the sender to resend   C) Corrects it automatically   D) Sends it to the next layer

158. Flow control is a technique for assuring that:

A) Data is encrypted properly   B) A transmitting entity does not overwhelm a receiving entity with data   C) Data is compressed   D) Routing is optimized

159. The simplest form of flow control is:

A) Sliding-window flow control   B) Stop-and-wait flow control   C) Go-back-N ARQ   D) Selective-reject ARQ

160. In stop-and-wait flow control, the source entity:

A) Sends all frames at once   B) Transmits a frame and waits for acknowledgment before sending the next   C) Sends multiple frames without waiting   D) Never waits for acknowledgment

161. In sliding-window flow control, the receiver allocates buffer space for how many frames?

A) 1 frame   B) W frames (window size)   C) All frames   D) No frames

162. The window size in sliding-window protocol represents:

A) The physical size of the cable B) The number of frames that can be sent without waiting for acknowledgment C) The total number of frames in the message D) The bandwidth of the link

163. The receiver's window in a sliding window protocol expands when:

A) An ACK is received B) An ACK is sent C) A frame is sent D) A frame is received

164. The sender's window in a sliding window protocol expands when:

A) An ACK is received B) An ACK is sent C) A frame is sent D) A frame is received

165. The stop-and-wait flow control method is the same as the sliding window method with a window size of:

A) 0 B) 1 C) 2 D) 3

166. The sender has a sliding window of size 15. The first 15 frames are sent. How many frames are in the window now?

A) 0 B) 1 C) 14 D) 15

167. In TCP, a SYN + ACK segment consumes how many sequence numbers?

A) 1 B) 2 C) 3 D) 4

168. Which event occurs during the transport layer three-way handshake?

A) Exchange of data B) Synchronization of sequence numbers C) Termination of connection D) Setting maximum window size only

169. HTTP uses which transport layer protocol?

A) UDP B) TCP C) Both D) Neither

170. DNS (user to DNS server) uses which transport layer protocol?

A) TCP only B) UDP C) Both equally D) Neither

171. RTP (video, voice, games) uses which transport layer protocol?

A) TCP B) UDP C) Both D) Neither

172. DHCP uses which transport layer protocol?

A) TCP B) UDP C) Both D) Neither

173. Which application uses UDP as the transport layer protocol?

A) HTTP B) POP3 C) DHCP D) SMTP

174. Transport layer uses _____ for session addressing.

A) IP address B) MAC address C) Port number D) Domain name

175. The address of HTTP application is:

A) Port 20,21 B) Port 25 C) Port 80 D) Port 110

176. The address of HTTPS application is:

A) Port 8080 B) Port 25 C) Port 21 D) Port 443

177. The address of SMTP application is:

A) Port 20 B) Port 25 C) Port 21 D) Port 23

178. The address of FTP application is:

A) Port 20,21 B) Port 22 C) Port 23 D) Port 25

179. The address of SSH application is:

A) Port 20 B) Port 22 C) Port 23 D) Port 25

180. The range of ports reserved for system services or commonly used protocols is:

A) 49152 - 65535 B) 1024 - 49151 C) 0 - 1023 D) 1 - 256

181. UDP packets have a fixed-size header of ______ bytes.

A) 8 B) 16 C) 32 D) 48

182. The length of the acknowledgment number field in TCP is ______ byte(s).

A) 1 B) 2 C) 3 D) 4

183. The length of the window size field in TCP is ______ byte(s).

A) 1 B) 2 C) 3 D) 4

184. The sizes of sequence number in TCP header is:

A) 8 bits B) 16 bits C) 32 bits D) 64 bits

185. Go-back-N ARQ is a form of error control based on:

A) Stop-and-wait flow control B) Sliding-window flow control C) Selective-reject protocol D) CRC only

186. In Go-back-N ARQ, when the destination detects an error in a frame, it:

A) Accepts the frame and corrects it B) Discards that frame and all future frames until the frame in error is correctly received C) Sends a positive acknowledgment D) Ignores the error

187. With selective-reject ARQ, the only frames retransmitted are:

A) All frames after the error B) Only those that receive a negative acknowledgment or time out C) All frames in the window D) No frames

188. RNR (Receive Not Ready) message in sliding-window protocol means:

A) The receiver is ready for all frames B) The receiver acknowledges former frames but forbids transfer of future frames C) The receiver has an error D) The connection is closed

189. Piggybacking in sliding-window protocol means:

A) Sending data in a separate frame B) Sending both data and acknowledgment together in one frame C) Sending multiple acknowledgments D) Not sending any acknowledgment

190. For a 3-bit sequence number field, frames are numbered modulo:

A) 4   B) 8   C) 16   D) 32